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Chapter 21

GELFAND-P001001

– · –

8/II-82 Gelfand Sem..
Müller Werner.
Lattice gravity. (work with Cheeger and Schräder)

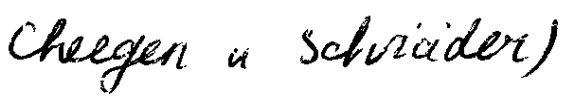


Diagram and typed description. The manuscript spells the name “Schräder”.

Original source details: GELFAND-P001001

1) Consider a one-dim. mech. system with Hamiltonian:

$$H(p,q) = +\frac{p^2}{2m} + V(q) \tag{1}$$

$q \to Q, \quad p \to P$ operators on $L^2(\mathbb{R})$

$$Qf(x) = xf(x), \qquad Pf(x) = i\bar{h}\frac{d}{dx}f(x)$$

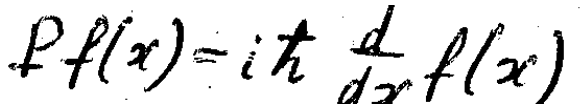


Diagram and typed description. There is no minus sign before $i\bar{h}$ in the expression for $Pf(x)$.

Original source details: GELFAND-P001001

$$H(p,q) \to \widehat{H}$$

Consider Schrödinger’s eq.

$$i\frac{\partial}{\partial t} = \bar{h}^{-1}\widehat{H}f \tag{2}$$

giving the evolution:

$$f(x,t) = e^{-i\frac{t}{\bar{h}}\widehat{H}}f(x,0) \tag{3}$$

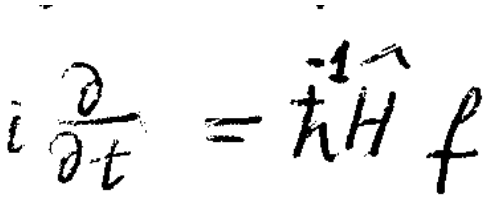


Diagram and typed description. On the left of (2) the manuscript has only $i\,\partial/\partial t$. The letter f is on the right.

Original source details: GELFAND-P001001

Prob. of transition from x to x' in time [added above the line: from t_0 to] t, according to Feynman

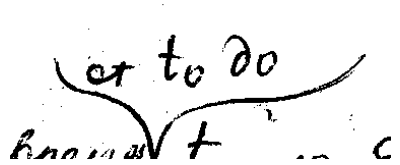


Diagram and typed description. The words “from t_0 to” are inserted before t.

Original source details: GELFAND-P001001

$$\langle x' \mid e^{-i\frac{t}{\hbar}\widehat{H}} \mid x\rangle = \int_{\substack{\text{over [inserted: all] paths } x \\ x(0)=x,\ x(t)=x'}} e^{-iS[x]} \prod_{\tau} \frac{dx(\tau)}{2\pi} \tag{4}$$

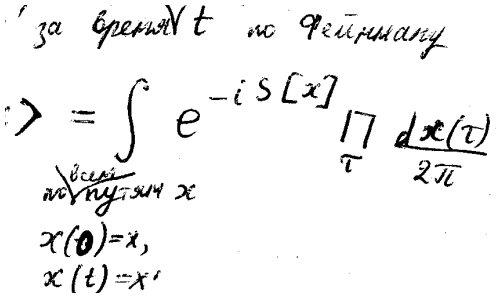
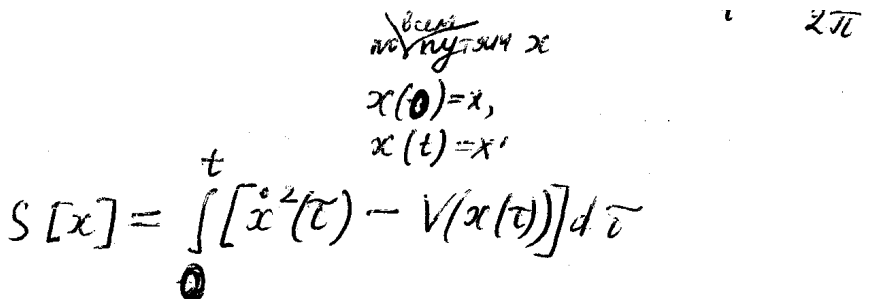


Diagram and typed description. The word “all” is inserted above “over paths”.

Original source details: GELFAND-P001001

$$S[x] = \int_0^t [\dot{x}^2(\tau) - V(x(\tau))]d\tau \tag{5}$$



Original source details: GELFAND-P001001

Diagram and typed description. The exponent in (4) is $-iS[x]$. In (5) there is no written coefficient before $\dot{x}^2(\tau)$.

(see last chapter of the book by Konopleva and Popov)



Uncertain reading. The Russian surname reads «Коноплевой», with no distinct dots above the «е».

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GELFAND-P001002

[source number: 4]

or: insert $t - t_0$ in place of t

in place of 0 — ditto — t_0

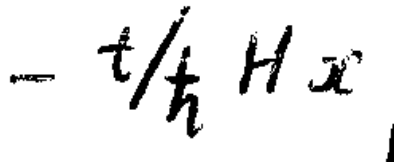
M. Kac gave some meaning to int. (4) with the substitution $i \to 1$

$$\overline{h}\frac{\partial f}{\partial t} = \widehat{H} f, \tag{6}$$

then

$$\langle x' \mid e^{-\frac{t}{\hbar} Hx} \mid x \rangle = \int_{\substack{\text{over all paths} \\ x:\, x(0)=x \\ x(t)=x'}} e^{-\int_0^t x(\tau) d\tau}\, dW(x) \tag{7}$$

[Arrow to $dW(x)$:] Wiener measure.



Uncertain reading. The letter x follows H. Its status as a subscript is unclear.

Original source details: GELFAND-P001002

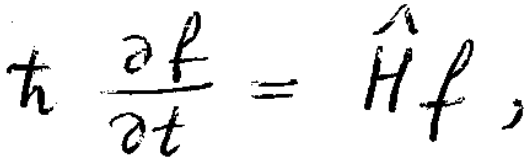


Diagram and typed description. In (6) there is no minus sign before $\widehat{H} f$ on the right.

Original source details: GELFAND-P001002

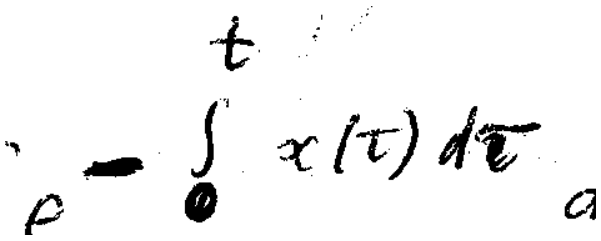


Diagram and typed description. The integrand in the exponent reads $x(\tau)$.

Original source details: GELFAND-P001002

2) Gravity.

$$i \mapsto 1 \qquad \rightsquigarrow$$

[Crossed out: “for” (?)] Lorentz. metric $\longmapsto$ Riemannian metric.



Uncertain reading. The crossed-out word before “Lorentz.” tentatively reads “for”.

Original source details: GELFAND-P001002

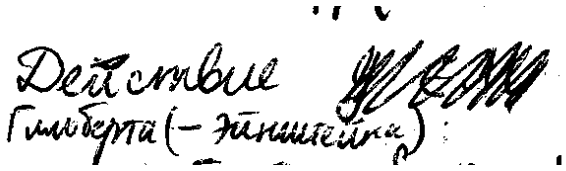
M — a compact C^∞ manifold

$$\mathcal{M} = \{\text{space of all Riem. metrics on } M\}$$

$$G = \mathrm{Diff}(M) \quad \text{— gauge group.}$$

Action [crossed out: illegible] [added below: Hilbert (– Einstein)]:

$$S[g] = \lambda \int_M R\, d\,\mathrm{vol}, \qquad R \text{ — scalar curvature} \tag{8}$$



Uncertain reading. The earlier entry after "Action" is crossed out. Below it is "Hilbert (– Einstein)".

Original source details: GELFAND-P001002

(nontrivial: Einstein thought there was no action)

— R contains 2nd derivs. ⇒ a priori a 4th-order eq.

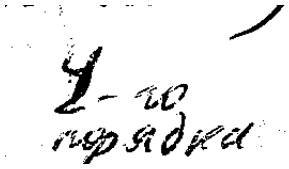
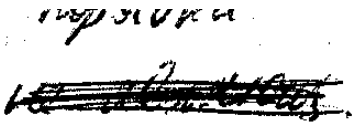


Diagram and typed description. The end of the line reads "4th order".

Original source details: GELFAND-P001002

But locally there is a noncovar. addition of divergence form, [crossed out: illegible] leaving S unchanged.



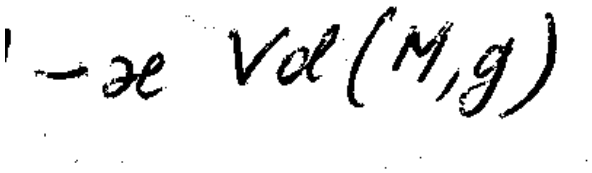
Uncertain reading. The reading under the cancellation after "form" is unresolved.

Original source details: GELFAND-P001002

Problem (I.M.G.) Find all comb. of curvatures yielding a lower-order eq.

"Answer": Einstein and Pontryagin classes. Action:

$$\int_{\mathcal{M}/G} e^{-S[g] - æ\,\mathrm{Vol}(M,g)} d\mu(g) \tag{9}$$

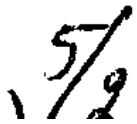


Uncertain reading. The coefficient before $\mathrm{Vol}(M,g)$ resembles "æ", possibly a handwritten $\varkappa$.

Original source details: GELFAND-P001002

Formally:

$$d\mu(g) = \prod_{x \in M} \det g(x)^{5/2} \prod_{\mu \le \nu} dg_{\mu\nu}(x) \tag{10}$$



Original source details: GELFAND-P001002

Diagram and typed description. The exponent reads $5/2$, with no minus sign.

(Faddeev – Popov)

Lattice approximation.

[A number is crossed out at left.]

$$P^n \subset \mathbb{R}^N \qquad P^n \underset{\varphi}{\longrightarrow} M \quad \text{triangulation}$$

[Stroke from P^n to the label:] piecewise linear subset.



Original source details: GELFAND-P001002

Uncertain reading. Tentative reading of the crossed-out entry “(11)”.

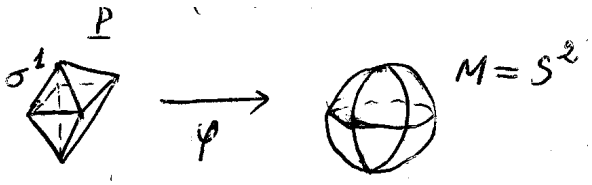
GELFAND-P001003

The metric on M induces a piecewise-lin. metr. on P.

$$(P, \ell), \qquad \ell : P^{(1)} \to \mathbb{R}_+$$

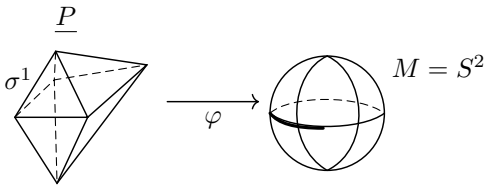
(simpl. $\rightsquigarrow$ its length)

(11)



Original source details: GELFAND-P001003

Diagram and typed description. Labels P, σ^1, φ and $M = S^2$. The dashed strokes follow the visible portions.



Discr. version of $\int_M R\,d\,\mathrm{vol}$.

$$P^n \longrightarrow M \qquad \text{triangulation} \tag{12}$$

$$\sigma^{n-2} \subset P^n$$

$\delta(\sigma^{n-2})$ — angle defect of σ^{n-2}

$$\sigma^n \supset \sigma^{n-2}$$

$[\sigma^{n-2}, \sigma^n]$ — dihedral angle

$$\delta(\sigma^{n-2}) = 2\pi - \sum_{\sigma^n \supset \sigma^{n-2}} [\sigma^{n-2}, \sigma^n] \tag{13}$$

$$R(P) = \sum_{\sigma^{n-2} \subset P} \delta(\sigma^{n-2})\,\mathrm{vol}(\sigma^{n-2}) \tag{14}$$

(Regge)

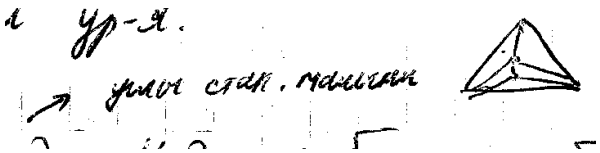
P_n — a seq. of triangulations, $P_n \to M$

$$\mathrm{mesh}(P_n) \underset{n\to\infty}{\longrightarrow} 0 \tag{15}$$

$$\text{Th.} \quad \text{Then} \quad R(P_n) \underset{n\to\infty}{\longrightarrow} \int_M R\,d\,\mathrm{vol} \tag{16}$$

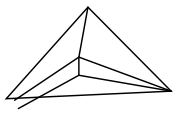
It is still unclear whether the eqs. converge.

Barycentric subdivision is unnecessary. [Upward arrow to a note:] angles become small.

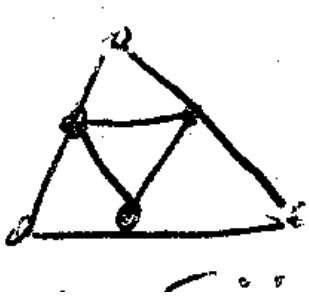


Original source details: GELFAND-P001003

Diagram and typed description. The interior strokes follow the visible joins. Some of the lower lines overlap.

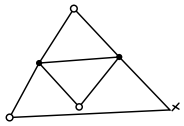


The angles must be bounded away from 0:



Original source details: GELFAND-P001003

Diagram and typed description. The marks at the vertices are reproduced as strokes. It is unclear whether they are letters.



a smooth triangulation is taken. (exists in Müller's work on Ray–Singer torsion)

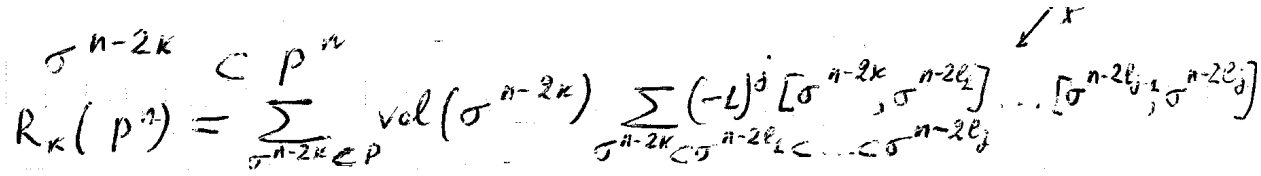
– · –

$$\int \operatorname{tr}(\Omega^k), \qquad \Omega \text{ — curvature tensor} \tag{17}$$

[Below Ω, with a stroke to it:] 2-form.

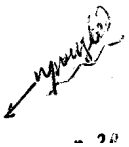
$$R_k(P^n) = \sum_{\sigma^{n-2k} \in P} \operatorname{vol}(\sigma^{n-2k}) \sum_{\sigma^{n-2k} \subset \sigma^{n-2\ell_1} \subset \cdots \subset \sigma^{n-2\ell_j}} (-1)^j \\ [\sigma^{n-2k}, \sigma^{n-2\ell_1}] \dots [\sigma^{n-2\ell_{j-1}}, \sigma^{n-2\ell_j}] \tag{18}$$

[Slanted above (18), with an arrow down and left:] norm. (?)



Original source details: GELFAND-P001003

Diagram and typed description. The last pair of indices in the product reads ℓ_{j-1}, ℓ_j.



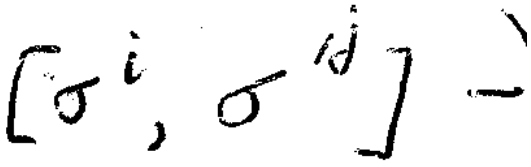
Uncertain reading. The note tentatively reads "norm.", with indistinct initial strokes.

Original source details: GELFAND-P001003

GELFAND-P001004

[source number: 5]

$[\sigma^i, \sigma^{ij}]$ — [inserted above: normalized] dihedral angle (solid) [Arrow to "solid":] normalized to 1.

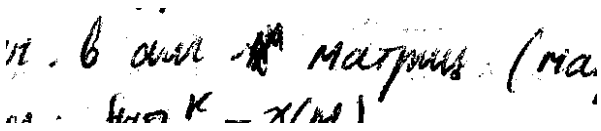


Uncertain reading. The second superscript reads ij, with a faint initial stroke.

Original source details: GELFAND-P001004

$$\underline{\text{Th.}} \quad R_k(P_n) \longrightarrow \int_M \operatorname{tr}(\Omega^k) \tag{19}$$

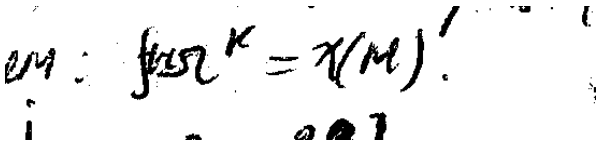
Ω — 2-form with values in the alg. [crossed-out sign] of matrices (a matrix of 2-forms).



Uncertain reading. Crossed-out sign, possibly A.

Original source details: GELFAND-P001004

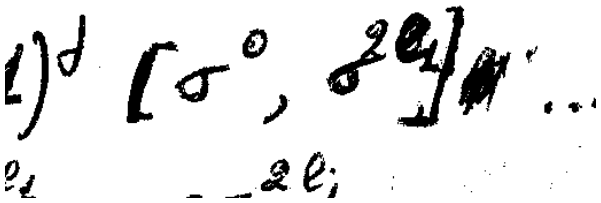
$$\underline{k = n/2} \quad \Rightarrow \quad \text{we obtain:} \quad \int \operatorname{tr} \Omega^k = \chi(M).$$



Uncertain reading. The manuscript reads $\int \operatorname{tr} \Omega^k = \chi(M)$.

Original source details: GELFAND-P001004

$$\chi(M) = \sum_{\substack{\sigma^0 \subset \sigma^{2\ell_1} \subset \cdots \subset \sigma^{2\ell_j} \\ \ldots [\sigma^{2\ell_{j-1}}, \sigma^{2\ell_j}]}} (-1)^j [\sigma^0, \sigma^{2\ell_1}] \tag{20}$$

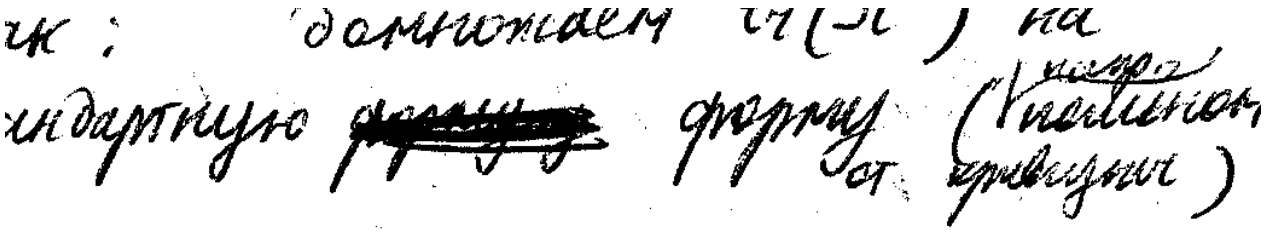


Uncertain reading. Illegible crossed-out sign after the first closing bracket.

Original source details: GELFAND-P001004

There is something similar in a work by Brin, presented at the seminar ca. 1946. It is unknown what it gives in the limit.

$\int_M \operatorname{tr}(\Omega^k)$ is def. as follows: multiply $\operatorname{tr}(\Omega^k)$ by the standard [crossed out: form?] form ([inserted above: inv.] polynomial in the curvature).



Original source details: GELFAND-P001004

Uncertain reading. The crossed-out word tentatively reads “form.”

$$\mathcal{M}(P^n) = \{\text{space of all piecewise-lin. metrics on } P^n\} \tag{21}$$

$\sigma^n \subset \mathbb{R}^N \longrightarrow n$-dim. simplex

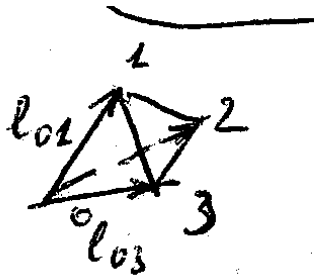
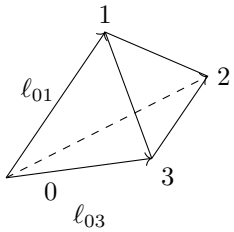


Diagram and typed description. Vertices $0, 1, 2, 3$, with edge labels ℓ_{01} and ℓ_{03}.



Original source details: GELFAND-P001004

$$\ell \rightsquigarrow g = (g_{ij}), \qquad g_{ij} = \frac{1}{2}(\ell_{0i}^2 + \ell_{0j}^2 - \ell_{ij}^2)$$

symm. and positive definite

$$\mathcal{P} = \{\text{symm. pos. def. matrices } n \times n\}$$

" $GL(n)/O(n)$

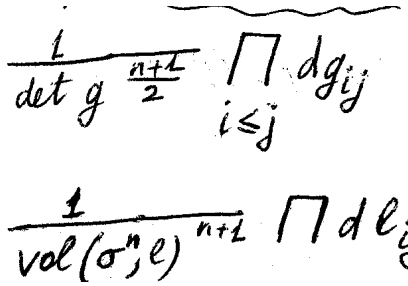
$$\mathcal{M}(\sigma^n) \equiv GL(n)/O(n)$$

Inv. measure on $\mathcal{M}(\sigma^n)$:

$$\frac{1}{\det g^{\frac{n+1}{2}}} \prod_{i \le j} dg_{ij} \tag{22}$$

$$\frac{1}{\mathrm{vol}(\sigma^n, \ell)^{n+1}} \prod d\ell_{ij} \tag{23}$$

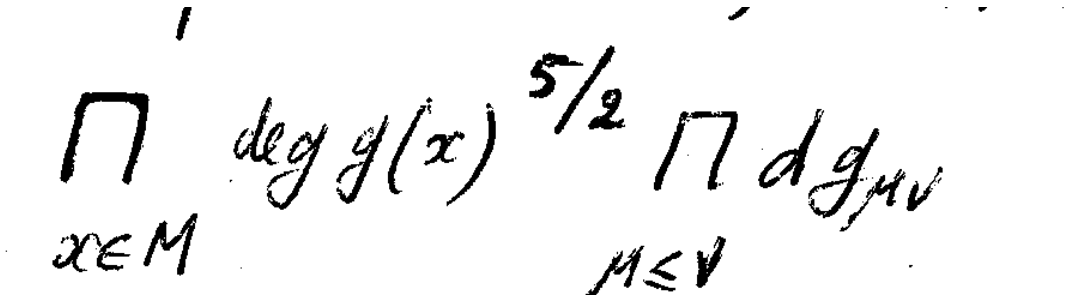
— measure on $\mathcal{M}(\sigma^n)$ (another way of writing it).



Uncertain reading. In (23) the manuscript gives $\prod d\ell_{ij}$.

Original source details: GELFAND-P001004

$$\prod_{x \in M} \deg g(x)^{5/2} \prod_{\mu \le \nu} dg_{\mu\nu} \tag{24}$$



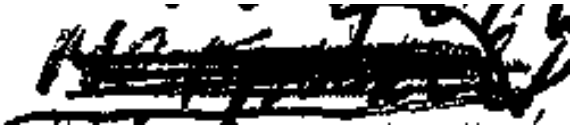
Original source details: GELFAND-P001004

Uncertain reading. The final stroke of "deg" descends below the line, unlike "det" in (22). The exponent in (24) is written as $5/2$.

$$\mathcal{M}(P^n) \hookrightarrow \prod_{\sigma^n \subset P^n} \mathcal{M}(\sigma^n) \tag{25}$$

(sub-set)

An induced [crossed out on the line: metric?] [circled and crossed out above: curvature?] [circled below: metric] arises on $\mathcal{M}(P^n)$ [crossed out on the right: not satisfy...].



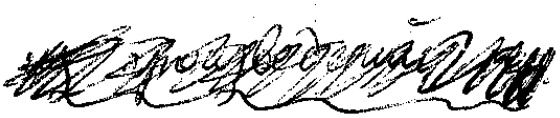
Original source details: GELFAND-P001005

Uncertain reading. On the line, tentatively "metric," with most of the word crossed out.



Original source details: GELFAND-P001005

Uncertain reading. The circled and crossed-out word above the line may read "curvature" or "metric."



Original source details: GELFAND-P001005

Uncertain reading. Under the cancellation, "not satisfy..." is legible. The ending and following strokes remain unclear.

($\rightsquigarrow$ gluing cond.)

A measure is introduced on the space of all metrizations. $\Rightarrow$ volume induced by the Riem. metric.

$$P_n \underset{n\to\infty}{\longrightarrow} M$$

(On $\mathcal{M}(\sigma^n)$ the inv. metric is the one on a symmetric space.)

Idea: $M^n \subset E^N$ isometric.

Chern's kinematic formula:

$$k = N - n + 2,$$

$\mathcal{G}_{N,k}$ = affine Grassmannian of $E^k \subset E^N$

dE^k — measure on $\mathcal{G}_{N,k}$

$$\int_{E^k \in \mathcal{G}_{N,k}} \chi(M^n \cap E^k)\, dE^k = c_{N,k} \int_M R\, d\,\mathrm{vol} \tag{26}$$

$$\int_{\mathcal{G}_{N,k}} \chi(P^n \cap E^k)\, dE^k = c_{N,k} R(P^n) \tag{27}$$

[Under P^n, with a vertical stroke:] simplicial sub-set

[Below the right-hand side:] (combinat. scalar curvature)

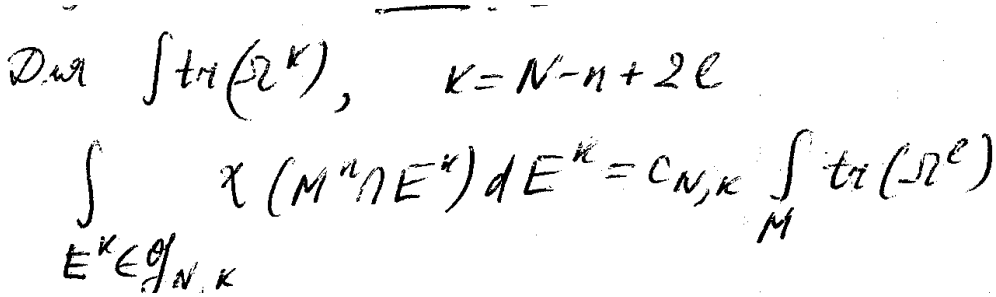
$$P_n \longrightarrow M$$

the integrands in (26) and (27) will agree a.e.

— . —

For $\int \operatorname{tr}(\Omega^k)$, $k = N - n + 2\ell$

$$\int_{E^k \in \mathcal{G}_{N,k}} \chi(M^n \cap E^k)\, dE^k = c_{N,k} \int_M \operatorname{tr}(\Omega^\ell) \tag{28}$$



Original source details: GELFAND-P001005

Uncertain reading. The line before (28) reads Ω^k, and the right-hand side of (28) reads Ω^ℓ.

$$\underset{\text{left-hand side of (27)}}{\text{— " —}} = c_{N,k} R_\ell(P^n). \tag{29}$$

GELFAND-P001006

[source number: 6]

15/II-82 Gelfand Sem.

Ginzburg.

M — sympl. manifold

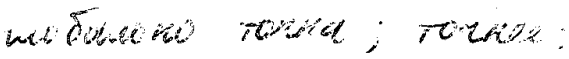
(ω — corresp. 2-form)

A — group acting on M in a Hamiltonian way.

1) A preserves ω

$\xi \in \mathfrak{A}$ — corresp. Lie algebra

$\xi \longrightarrow \omega(\xi, \cdot)$ globally (?) exact; exact:



Original source details: GELFAND-P001006

Uncertain reading. The word before "exact" tentatively reads "globally," with a tall medial loop. "Locally" is also possible.

2) $\mathfrak{A} \ni a \longmapsto H_a$ (Hamiltonian)
homomorphism of Lie algebras.
moment map: $\mu\colon M \longrightarrow \mathfrak{A}^*$

$$m \longmapsto \mu(m), \qquad \mu(m)(\underset{\uparrow}{a}) = H_a(m)$$
$$\mathfrak{A}$$



Original source details: GELFAND-P001006

Diagram and typed description. The arrow from $\mathfrak{A}$ below points to the argument a.

Commutes with the action of A and is compat. with the Poisson bracket.

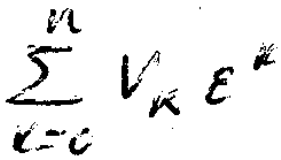
[Two short slanted strokes.]

GELFAND-P001007

22/II-82 Gelfand Sem.
Arnold.

$$V(\varepsilon) = \sum_{i=0}^{n} V_k \varepsilon^k, \qquad \sigma_i(k) = \sum_{C_n^i} k_1 \dots k_i$$

[Under $V(\varepsilon)$:] volume (ε-neighborhood of a conv. body). (Minkowski's theorem — this is a polynomial.)



Uncertain reading. The summation starts at $i = 0$, while the summand reads $V_k \varepsilon^k$.

Original source details: GELFAND-P001007

$$k_s \text{ — eigenvalues of } \frac{2 \text{ quadr. f.}}{1 \text{ quadr. f.}}$$

$$V_i = C_i \int_{\partial D} \sigma_i(k)\, dS$$

[Short vertical stroke below C_i.]

$$V_k = \widetilde{C}_k \underbrace{\int_{\xi \in G_{n,k}} \text{mes}(\pi_\xi D)\, d\xi}_{\text{"}}, \qquad \pi_\xi : \mathbb{R}^n \longrightarrow \xi$$

$$\int_{H_{n,k}} \chi(h \cap D)\, dh$$

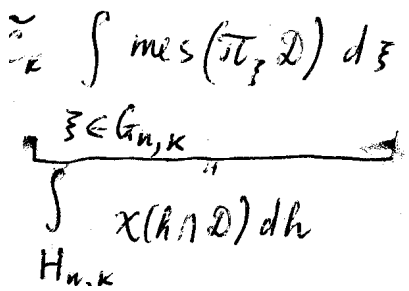


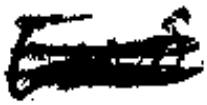
Diagram and typed description. A bracket with turned-up ends and two short slanting strokes appear below the upper integral.

Original source details: GELFAND-P001007

$H_{n,k}$ — all planes, not necessarily passing through 0

χ — Euler char..

This is the method of [crossed out: Buff...] Buffon (later generalized by Santaló) [Below "method of Buffon":] for computing the number π.



Uncertain reading. The first strokes, "Buff...", are visible under the cancellation.

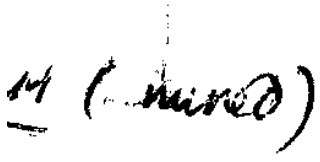
Original source details: GELFAND-P001007

Computation of the constants C_k, $\widetilde{C}_k$ — using balls.

This carries over to the lattice case.

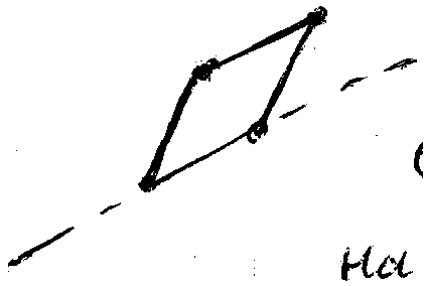
In place of volume, the number of integer points.

integral parallelogram (...piped?)



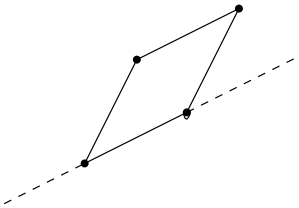
Uncertain reading. The letters after the dots tentatively read "piped."

Original source details: GELFAND-P001007



Original source details: GELFAND-P001007

Diagram and typed description. The lower edge continues as a dashed line. A loop at the lower right vertex may be a separate 0.



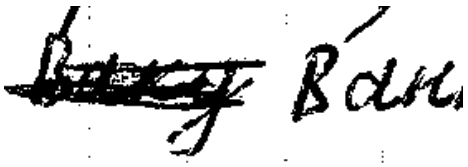
The volume of a parallelepiped is the number showing how many elem. par.-pipeds there are.

A lattice of intersections arises on the line.

Take the quotient space and this gives an analogue of height.

(there is an analogue of a lattice there too)

Mac Mullen a talk in [crossed out: Vancou...] Vancouver — there are refs. there.



Original source details: GELFAND-P001007

Uncertain reading. The first spelling is crossed out. The beginning reads “Vancou...”.

I. M. Gelfand.

On Mc Donald’s book “Symmetric Functions and Hall Polynomials”

Problem. Consider a partition (partition)

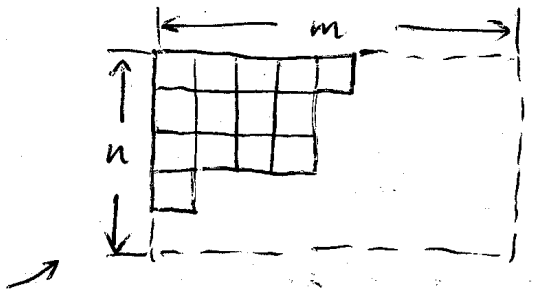
$$(L) \qquad \lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_k \qquad \text{(integers)}$$

Adding zeros is allowed.

Ways of writing it: (L) or

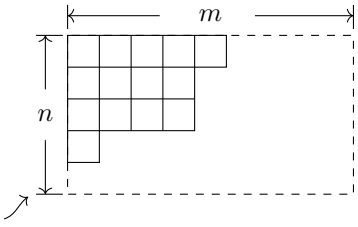
$$\begin{array}{ll} 5 & \bullet\ \bullet\ \bullet\ \bullet\ \bullet \\ 4 & \bullet\ \bullet\ \bullet\ \bullet \\ 4 & \bullet\ \bullet\ \bullet\ \bullet \\ 1 & \bullet \end{array}$$

Squares are better than dots (Young diagrams).



Original source details: GELFAND-P001007

Diagram and typed description. Rows of $5, 4, 4, 1$ cells in a dashed frame marked m and n.



Place it in a rectangle of dimensions $m \times n$ and consider the numbers

GELFAND-P001008

[source number: 11]

$$(*) \quad \begin{cases} \lambda_i + n - i, & 1 \le i \le n, \\ n - 1 + j - \lambda'_j, & 1 \le j \le m \end{cases}$$

$\lambda' = 4, 3, 3, 3, 1$ — dual diagram

Problem:

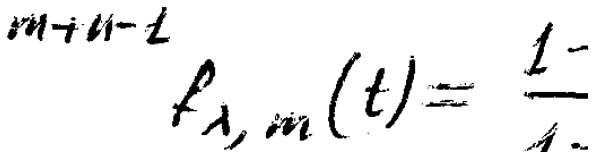
The list of $m + n$ numbers $(*)$ — is a permutation of the numbers $0, 1, 2, \ldots, m + n - 1$.

Corollary.

$$f_{\lambda,n}(t) \stackrel{\text{def}}{=} \sum_{i=1}^{n} t^{\lambda_i + n - i}$$

Then

$$f_{\lambda,n}(t) + t^{m+n-1} f_{\lambda,m}(t) = \frac{1 - t^{m+n}}{1 - t}$$



Uncertain reading. The second term reads $f_{\lambda,m}(t)$, with no visible prime on λ and no inverse argument.

Original source details: GELFAND-P001008

I. M. Gelfand from the work of Guillemin



Uncertain reading. The manuscript surname tentatively reads “Гийемина” (Guillemin).

Original source details: GELFAND-P001008

New cond. for Weyl’s asympt. formula

Sympl. cone: $\rho_a^* \omega = a\omega, \quad a > 0.$

Example 1. $\mathbb{R}^{2n} = \{(x, \xi)\}$, $x \in \mathbb{R}^n$, $\xi \in (\mathbb{R}^n)^*$, $\rho_a(x, \xi) = (a^{1/2}x, a^{1/2}\xi)$.

Example 2. $T^*X \qquad a(x, \xi) = (x, a\xi)$.



Uncertain reading. The base after T^* is heavily overwritten and tentatively reads X.

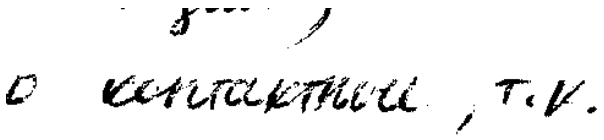
Original source details: GELFAND-P001008

Arnold.

Locally all cones are the same: the quotient is a contact manifold.

(Sympl. cones $\Longleftrightarrow$ contact manifolds)

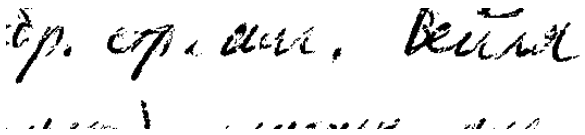
The quantization must be contact quantization, since we need the action — a 1-form!!



Uncertain reading. The ending of the word following "must be" tentatively gives the singular "contact."

Original source details: GELFAND-P001008

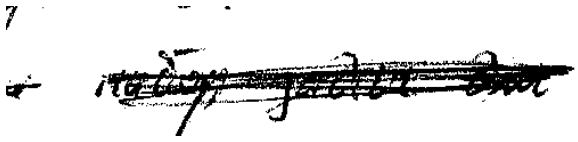
Claim. Can always be quantized and the Weyl alg. is constr. uniquely, microlocally unique, and topol. (microlocally) classical Weyl alg.



Uncertain reading. The abbreviation before "Weyl alg." reads "стр." (constr.).

Original source details: GELFAND-P001008

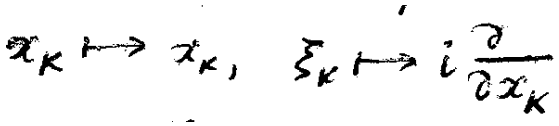
Quantization — [crossed out: a set...]



Uncertain reading. The beginning of the crossed-out line reads "a set...".

Original source details: GELFAND-P001008

$$x_k \longmapsto x_k, \qquad \xi_k \longmapsto i\frac{\partial}{\partial x_k}.$$



Uncertain reading. The image of ξ_k contains i with no minus sign.

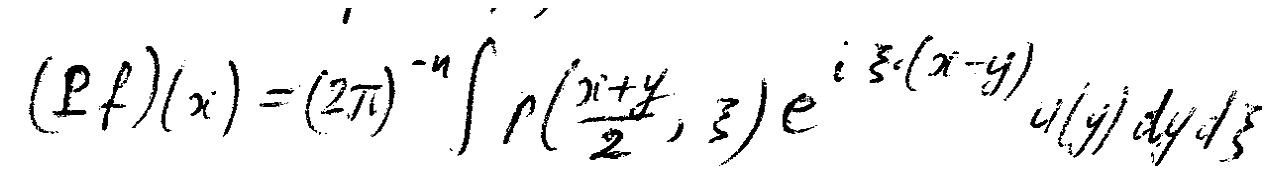
Original source details: GELFAND-P001008

$$p(x,\xi) = \sum_{i=0}^{\infty} p_i(x,\xi), \qquad p_i \text{ — homog. of degree } 2\mu - 2i$$

$$\rho_a(x,\xi) = (a^{1/2}x, a^{1/2}\xi); \qquad \mu \in \mathbb{C}.$$

(This is a symbol of order μ).

$$(Pf)(x) = (2\pi)^{-n} \int p\left(\frac{x+y}{2}, \xi\right) e^{i\xi\cdot(x-y)} u(y)\, dy\, d\xi$$



Original source details: GELFAND-P001008

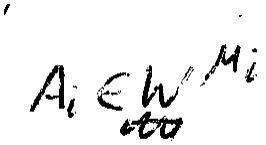
Uncertain reading. The left-hand side reads $(Pf)(x)$ and the integrand uses $u(y)$.

W^μ — space of such operators, $\underline{W^{\mu-1} \subset W^\mu}$.

Symbol: element of $W^\mu / W^{\mu-1}$.

H — Hilbert space, S — dense subspace on which all such op. act.

$\underline{W^\mu}$ — corresp. space of op. (of the form $A_1 + \cdots + A_k$, $A_i \in W^{\mu_i}$ in the example).



Uncertain reading. Below W are crossed-out looped strokes. The subscript reading is unresolved.

Original source details: GELFAND-P001008

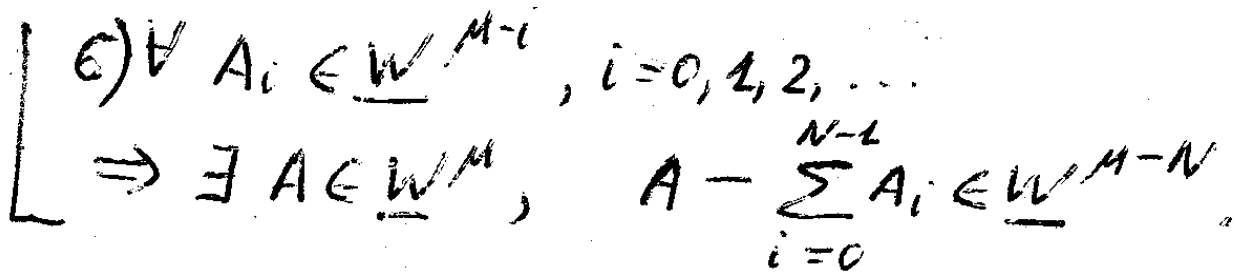
Axioms:

$$\underset{A}{\text{group}} \quad \begin{cases} 1) & \underline{W^\mu} \supset \underline{W^{\mu-1}} \\ 2) & A \in \underline{W^\mu}, \quad B \in \underline{W^\nu} \implies AB \in \underline{W^{\mu+\nu}} \\ 3) & A \in \underline{W^\mu}, \quad B \in \underline{W^\nu} \implies [A,B] \in \underline{W^{\mu+\nu-1}} \\ 4) & A \in \underline{W^\mu} \implies A^* \in \underline{W^{\bar{\mu}}} \end{cases}$$

$$5) \quad I \in \underline{W^0}$$

GELFAND-P001009

$$\left| \begin{array}{ll} 6) & \forall A_i \in \underline{W^{\mu-i}}, \quad i = 0,1,2,\ldots \\ & \implies \exists A \in \underline{W^\mu}, \qquad A - \sum_{i=0}^{N-1} A_i \in \underline{W^{\mu-N}}. \end{array} \right.$$



Original source details: GELFAND-P001009

Diagram and typed description. To the left of item 6) is the lower end of a square bracket.

Group $\underline{B}$.

1°. If $\operatorname{Re}\mu \leq 0$, then $\forall A \in \underline{W^\mu}$ is bounded.

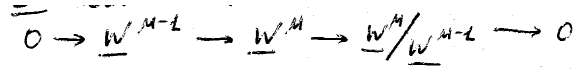
2°. If $\operatorname{Re}\mu < 0$, then $\forall A \in \underline{W^\mu}$ is compact.

3°. $\exists s_0 > 0$, if $A \in \underline{W^\mu}$, $\operatorname{Re}\mu < -s_0$; then A is nuclear.

Group $\underline{C}$.

$$A \in \underline{W^0}, \quad A \text{ is compact} \implies A \in \underline{W^{-1}}$$

$\underline{D}$. Exact seq.



Original source details: GELFAND-P001009

Diagram and typed description. Underlines on W and arrows of the sequence.

$$0 \longrightarrow \underline{W^{\mu-1}} \longrightarrow \underline{W^\mu} \longrightarrow \underline{W^\mu}/\underline{W^{\mu-1}} \longrightarrow 0$$

"Nicely" decomp. into a direct sum.

$E.$ $\exists A(\mu) \in \underline{W}^{\mu}$, depending analyt. on $\mu \in \mathbb{C}$, such that $A(0) - I \in \underline{W}^{-1}$ and

$$A(\mu)A(\nu) - A(\nu+\mu) \in \underline{W}^{\mu+\nu-1}.$$

GELFAND-P001010

[source number: 15]

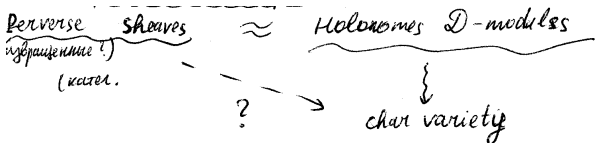
1/III-82. Gelfand Sem.

R. MacPherson. The title of a paper [crossed out: which?] he would speak about:



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Uncertain reading. The crossed-out English word tentatively reads “which.”



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Diagram and typed description. “Holonomes” and “char variety” retain the manuscript spellings. The wavy and dashed arrows point toward the lower label.

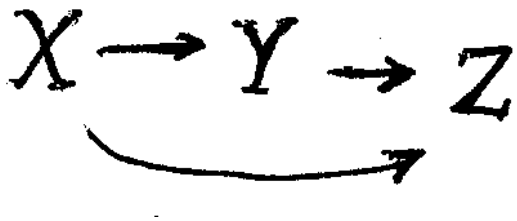
Perverse Sheaves (perverted?) (categ. ≈ Holonomes $\mathcal{D}$-modules

? char variety

Another title:

$H(X \longrightarrow Y)$ homol. theory of maps of spaces.

$$H(X \longrightarrow pt) = H_*(X)$$
$$H(X \xrightarrow{\text{id}} X) = H^*(X)$$



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Diagram and typed description. The lower arc points from X to Z.

$X \longrightarrow Y \longrightarrow Z$

there is a product, covering comp. of maps.

D. McDuff. Cohom. of the group of volume-preserving diffeom.

$$\mathcal{G} = \mathrm{Diff}^{c}_{\omega 0} M,$$

ω — volume form on M, M — a manifold;

c — with comp. support; 0 — connected component of Id.

$\mathcal{G}$ with the usual topology (C^∞).

G — the same group with the discr. top.

$$G \underset{\text{ident.}}{\longrightarrow} \mathcal{G}$$

$$\bar{B}\mathcal{G} \longrightarrow \overset{K(G,1)}{\overset{\shortparallel}{BG}} \longrightarrow B\mathcal{G} \qquad \text{(classifying spaces)}$$

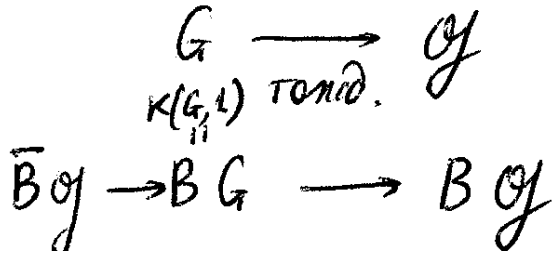


Diagram and typed description. Above BG is $K(G,1)$ with two short strokes. A stroke from $\bar{B}\mathcal{G}$ leads to the fiber annotation.

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homot. fiber (turn the map into a fibration, then $\bar{B}\mathcal{G}$ — fiber)

$\bar{B}\mathcal{G}$ depends only on a neighb. of 1 in $\mathcal{G}$.

This is a topological version of the Lie alg. in some sense.

Fuks:

$$G/H \longrightarrow BH \longrightarrow BG$$

(usually); if H is [inserted above: very] dense in G, then the fund. domain is an "infinitesim." neighb. of the identity.

$\bar{B}\mathcal{G} \times M$ has a foliation $\mathcal{F}$, transverse to all fibers $x \times M$ and defined by [inserted above: such] a closed n-form Ω, that $\Omega|_{x \times M} = \omega$.



Uncertain reading. The end of "closed" has heavily overwritten strokes.

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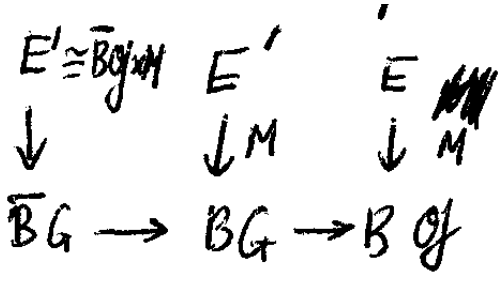


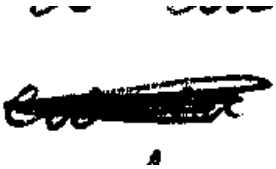
Diagram and typed description. The lower left label reads $\bar{B}G$ and the upper left uses $\bar{B}\mathcal{G}$. The product sign at top runs into the strokes of M. An entry above the right-hand M is heavily crossed out.

$$\begin{array}{ccccc} E' \cong \bar{B}\mathcal{G} \times M & & E' & & E \\ \downarrow & & \downarrow M & & \downarrow M \\ \bar{B}G & \longrightarrow & BG & \longrightarrow & B\mathcal{G} \end{array}$$

Original source details: GELFAND-P001010

E — std. bundle with fiber M.

E' — its lift, [crossed out: connection?] it has zero curvature, since the fiber of the corresp. principal bundle is discrete.



Original source details: GELFAND-P001010

Uncertain reading. The crossed-out word has an initial “c” and tentatively reads “connection.”

On the left, the restriction to the fiber becomes triv. by the property of the classifying space.

The zero-curvature connection is transferred and gives a foliation on $\bar{B}\mathcal{G} \times M$

GELFAND-P001011

$\Omega - \pi^*\omega = 0$ near ∞ on M.

Classes.

$$c_k(M) \in H^k\left(\bar{B}\,\mathrm{Diff}^c_{\omega_0} M; \underbrace{H_c^{n-k}(M;\mathbb{R})}_{\text{coefficients}}\right)$$

$n = \dim M$.

Let K — k-cycle in $\bar{B}\,\mathrm{Diff}^k_{\omega_0} M$.



Original source details: GELFAND-P001011

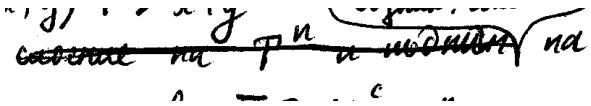
Uncertain reading. In the definition of the cycle the superscript on Diff is k. The neighboring formulas use c.

$$c_k(M)(K) = \text{class of the closed } (n-k)\text{-form } x \longmapsto \int_{K\times x} \Omega$$

Example. $M = T^n$, $K = T^n$.

$$T^n \times T^n \longrightarrow T^n$$
$$(x, y) \longmapsto x + y$$

Take [crossed out: a foliation on T^n and lift it?] [inserted above in a callout: a foliation arises] on $T^n \times T^n$. Obtain a foliation repr. a cycle in $\bar{B}\,\mathrm{Diff}^c_{\omega_0} T^n$.



Original source details: GELFAND-P001011

Uncertain reading. The crossed-out verb means “lift.” Its tense is uncertain.

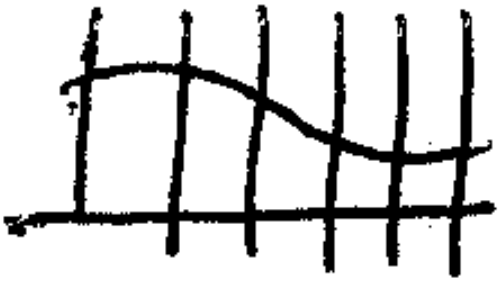
$$c_n(T^n)(T^n) = \int_{T^n \times x} \Omega \qquad \text{indep. of the choice of } x.$$

Prop. 1

$$\begin{aligned} c_n(S^n) &= 0 \quad \text{for } n \text{ even} \\ &\neq 0 \quad \text{for } n \text{ odd} \end{aligned}$$

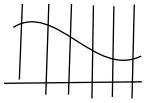
1st claim: $K^n \times S^n$ (K^n — any manifold)

Ω — form defining a foliation transv. to the spheres.



Original source details: GELFAND-P001011

Diagram and typed description. Six nearly vertical strokes, a transverse straight line and a curve.



Then $\int_{\substack{\text{horiz.} \\ \text{leaf}}} \Omega = 0.$

Constr. another class s

Prop. 2

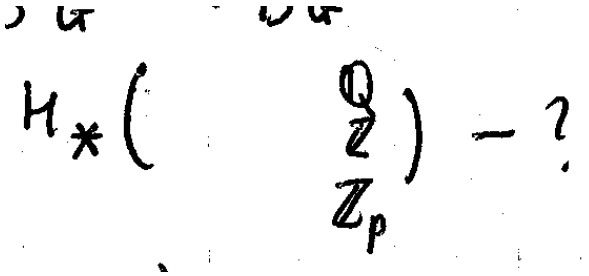
$$\begin{aligned} s(\mathbb{R}^n) &\neq 0 \quad n \text{ even} \\ &= 0 \quad n \text{ odd} \end{aligned}$$

Milnor.

G — Lie group, G^δ — the same with the discr. topology.

$$G^\delta \longrightarrow G$$
$$BG^\delta \longrightarrow BG$$

$$H_*\begin{pmatrix} \mathbb{Q} \\ \mathbb{Z} \\ \mathbb{Z}_p \end{pmatrix} - ?$$



Original source details: GELFAND-P001011

Uncertain reading. The space to the left of the column $\mathbb{Q}, \mathbb{Z}, \mathbb{Z}_p$ in the parentheses is blank.

$H_*(BG)$ are well known and well behaved

$H_*(BG^\delta)$ — ?

are unknown and badly behaved. It is unclear whether they are useful.

In algebraic K-theory, in foliation theory, in Hilbert's 3rd problem

— all this comes up. (equidecomposability)

GELFAND-P001012

[source number: 16]

Isomorphism conjecture:

$$(1) \qquad H_*(BG^\delta, \mathbb{Z}/p) \xrightarrow{\simeq} H_*(BG, \mathbb{Z}/p); \qquad \forall \text{ prime } p.$$

Applied to Hilb.'s 3rd problem by H. Sah, Dupont.

When are they equidecomposable? A partial answer comes from the cohomology of the classifying space.

Probably (1) is a very difficult assertion.

Chern–Weyl: G compact $\Rightarrow$

$$\widetilde{H}_*(BG^\delta, \mathbb{Q}) \xrightarrow{0} \widetilde{H}_*(BG, \mathbb{Q}).$$

(the map is trivial)

Sah–Wagoner:



Uncertain reading. The second surname reads "Wagoner."

Original source details: GELFAND-P001012

$$\underset{\text{connected}}{G} \neq 1 \quad \Rightarrow \quad \exists \text{ an epimorphism}$$

$$H_2(BG^\delta, \mathbb{Z}) \longrightarrow \underset{\substack{\text{uncount.}\\ \text{sum}}}{\bigsqcup} \mathbb{Q} \longrightarrow 0$$

At the same time:

$H_2(BG, \mathbb{Z})$ is finitely generated.

Let us discuss evid. for (1).

① (1) holds in the spec. case when the Lie alg. $\mathfrak{g}$ is solvable.

The proof is elementary.

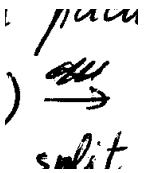
$$\bar{B}G \longrightarrow BG^\delta \longrightarrow BG$$

[Below $\bar{B}G$ with a vertical stroke:] prove by ind. that the cohom. of $\bar{B}G$ is trivial.

② Let G have finitely many components.

Then (1) is an epim. and splits.

$$H_*(BG^\delta, \mathbb{Z}/p) \underset{\text{split}}{\longrightarrow} H_*(BG, \mathbb{Z}/p) \longrightarrow 0$$



Uncertain reading. Short strokes above the arrow are crossed out, possibly "ep."

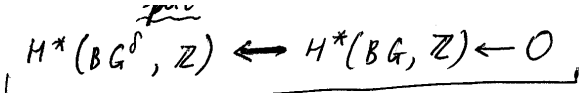
Original source details: GELFAND-P001012

[Crossed-out sign at left.] (2) Corollary.



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Uncertain reading. The crossed-out looped sign to the left of (2) is unresolved.



Original source details: GELFAND-P001012

Diagram and typed description. Both heads are visible on the middle arrow. A bracket appears below the expression.

$$H^*(BG^\delta, \mathbb{Z}) \longleftrightarrow H^*(BG, \mathbb{Z}) \longleftarrow 0$$

Gen. — thm. [crossed out: Du?] Bott–Heitsch.



Original source details: GELFAND-P001012

Uncertain reading. The cancellation before Bott–Heitsch may read “Du.”

③ (1) always holds [crossed out: when] for H_1 and in interesting cases for H_2:

$$\mathrm{SL}_n \mathbb{R}, \quad \mathrm{SL}_n \mathbb{C}.$$



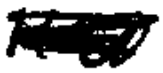
Original source details: GELFAND-P001012

Uncertain reading. A sign below the line between the index n and $\mathbb{R}$ is heavily crossed out, possibly $\mathbb{Z}$.

Haefliger.

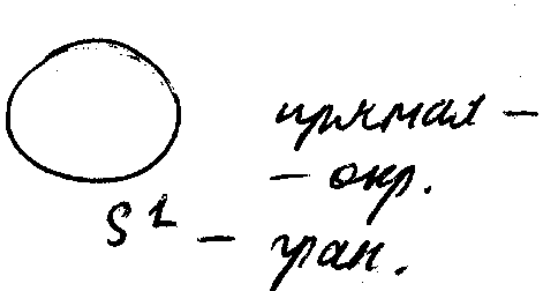
$\mathrm{PSL}_2(\mathbb{Z})$ on $\mathrm{Im}\, z > 0$.

(fractional-lin. transf. [crossed out: co?] with integral coeff.)



Original source details: GELFAND-P001012

Uncertain reading. The crossed-out fragment may begin “co.”



Original source details: GELFAND-P001012

Diagram and typed description. Circle and boundary label.

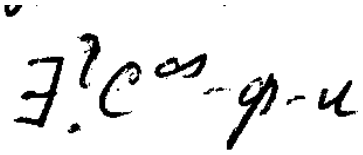


line —
— circ.
S^1 — bdry.

γ_1, γ_2 — generators.

$$\gamma_1 : z \longmapsto z + 1; \qquad \gamma_2 : z \longmapsto -\frac{1}{z}.$$

$\forall f \in C^\infty(S^1), \quad \exists?\, C^\infty$-fns. $f_1, f_2 \in C^\infty(S^1)$



Uncertain reading. A question mark with a separate dot follows $\exists$.

Original source details: GELFAND-P001012

GELFAND-P001013

$$(1) \qquad f(x) = f_1(x) - f_1(\gamma_1(x)) + f_2(x) - f_2(\gamma_2(x)).$$

This is related to foliations on a Riem. manifold whose leaves are min. surfaces.

Unknown!

Approximately true: approximation is possible (in the C^∞ top.) — this is easy.

GELFAND-P001014

15/III-82. Gelfand Sem.

Topics (for term papers, diploma theses, etc.)

Kirillov:

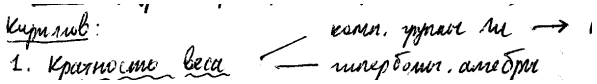
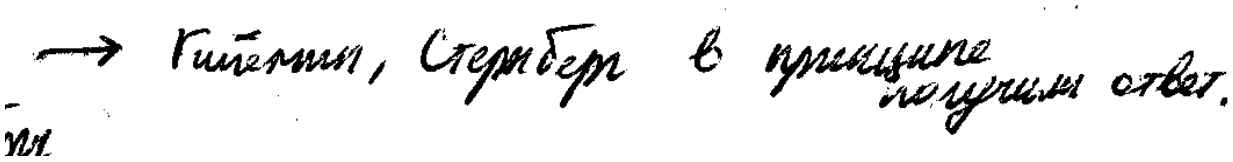


Diagram and typed description. The upper and lower strokes lead to two separate items.

1. Weight multiplicity — comp. Lie groups → Guillemin and Sternberg obtained an answer in principle.
   — hyperbolic algs.



Original source details: GELFAND-P001014

Diagram and typed description. The arrow after “comp. Lie groups” points to Guillemin and Sternberg.

Refs.: Macdonald (article)

Example problem in 2: a contragredient Lie algebra with a hyperbolic Cartan matrix (having a neg. eigenvalue) is considered.

Consider the action of $SU(3)$ on its Lie algebra:

$$X = \begin{pmatrix} i\varphi_1 & a & c \\ -\bar{a} & i\varphi_2 & b \\ -\bar{c} & -\bar{b} & i\varphi_3 \end{pmatrix}, \qquad \varphi_1 + \varphi_2 + \varphi_3 = 0$$

$$g \in SU(3) \Rightarrow g : X \to gXg^*.$$

GELFAND-P001015

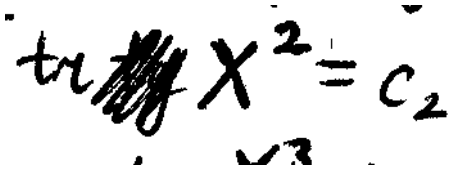
Consider the orbit Ω. [separate stroke] It has the form:



Uncertain reading. A separate slanted stroke follows the period. It could be 1 or part of an exclamation mark.

Original source details: GELFAND-P001015

$$\begin{cases} \text{tr [crossed-out sign] } X^2 = c_2, \\ \text{tr } X^3 = c_3 \end{cases} \qquad (\text{and, incidentally, } \operatorname{tr} X = 0)$$



Uncertain reading. Between tr and X^2 is a heavily crossed-out sign.

Original source details: GELFAND-P001015

(these traces recover the roots, and $\forall$ matr. can be brought into diag. form)

Denote it by Ω_{c_2,c_3}.

The orbit is of dimension 6 (gen. position).

Project onto the plane (2-dim.) $(\varphi_1, \varphi_2, \varphi_3)$, $\varphi_1 + \varphi_2 + \varphi_3 = 0$.

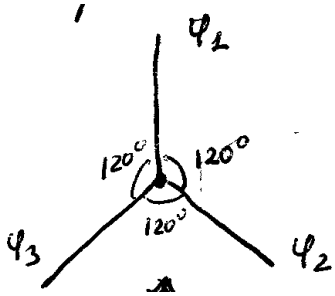
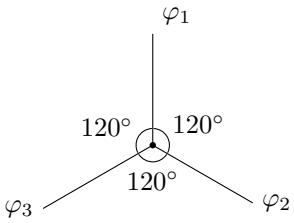


Diagram and typed description. Three axes labelled $\varphi_1, \varphi_2, \varphi_3$ with three $120°$ angles.



Original source details: GELFAND-P001015



Diagram and typed description. An upward arrow is visible in the crossed-out sketch. The underlying geometry and labels are unresolved.

Original source details: GELFAND-P001015

— project onto the plane $\varphi_1 + \varphi_2 + \varphi_3 = 0$ [crossed out: illegible] the coordinate axes.



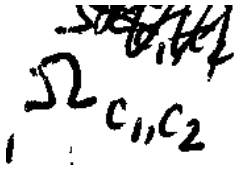
Uncertain reading. A short entry before "coordinate axes" is crossed out.

Original source details: GELFAND-P001015

— this is how it should be drawn. (p — projection)

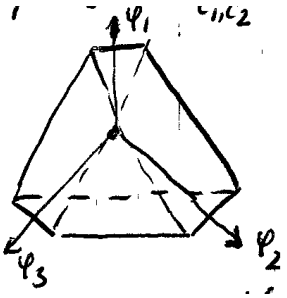
Facts:

a) The projection of Ω_{c_1,c_2} fills a hexagon.



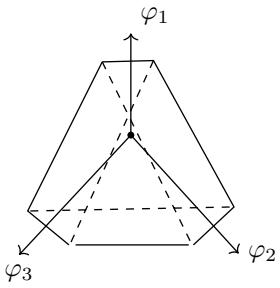
Original source details: GELFAND-P001015

Uncertain reading. The indices on this line are c_1, c_2. The designation above has c_2, c_3.



Original source details: GELFAND-P001015

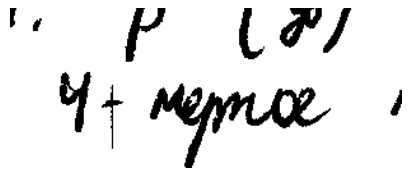
Diagram and typed description. Hexagon, three outgoing arrows and three dashed interior lines.



(this is the set of diag. entries of matrices with fixed eigenvalues $\lambda_1, \lambda_2, \lambda_3$)

b) Consider $p^{-1}(y_0)$, y_0 — a point of the hexagon.

It is a 4-dimensional manifold. Regularity fails on the dashed lines.



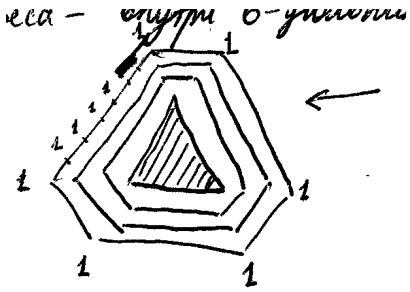
Original source details: GELFAND-P001015

Uncertain reading. A long thin vertical line crosses the horizontal stroke after 4. "4-dimensional" is the probable reading.

This is all in the paper of Berezin and Gelfand in Tr. MMO, 1956. ("Some remarks on the theory of spherical fns.")

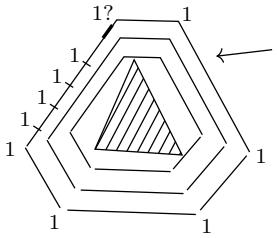
A rep. of $SU(3)$ is given by a pair of integers and the weights follow from them.

The weights are inside the hexagon. Multiplicities:



Original source details: GELFAND-P001015

Diagram and typed description. Three nested hexagonal contours surround a hatched triangle. Overlapping strokes obscure some marks at the upper-left corner.



$\leftarrow$ lines of changing multiplicity

$SU(3)$ — in Gelfand and Berezin.

The answer of Guillemin, Sternberg — for large c_2, c_3.

[Above the names, with an arrow to them:] preprint $\approx$ 1981

X — the preimage of *? a point, $X = p^{-1}(y_0)$.



Uncertain reading. A star-like sign between "preimage" and "point" may be a crossed-out entry.

Original source details: GELFAND-P001015

$\langle e^B \tau, X \rangle$ — multiplicity!!

[Below e^B the strokes of an equals sign are turned vertically.]

$$e^B = 1 + B + \frac{B \wedge B}{2} + \dots$$

B — sympl. form on the orbit.

τ — Todd class of the manifold.

I. M. G. Main question: duality between groups and their representations.

$$x^{\textcircled{n}} = x(x-1)\dots(x-n+1)$$

$$\textcircled{\frac{d}{dx}} f(x) = f(x+1) - f(x)$$

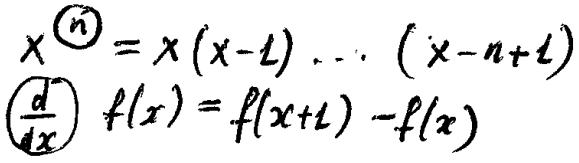


Diagram and typed description. The circles around n and $\frac{d}{dx}$ are part of the author's notation.

Original source details: GELFAND-P001015

GELFAND-P001016

[source number: 25]

Then

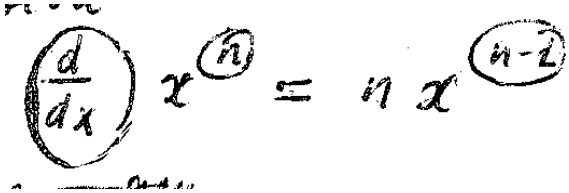


Diagram and typed description.

$\textcircled{\frac{d}{dx}} x^{\textcircled{n}} = n\, x^{\textcircled{n-1}}$

Original source details: GELFAND-P001016



Uncertain reading. A short entry is enclosed and crossed out. Its reading remains unresolved.

Original source details: GELFAND-P001016

Dual: for a representation, look at the weight multiplicities.

There is a (unique) invariant volume on the orbit.

Normalization: volume of the orbit = dimension of the linear representation associated with it.

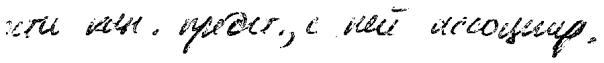
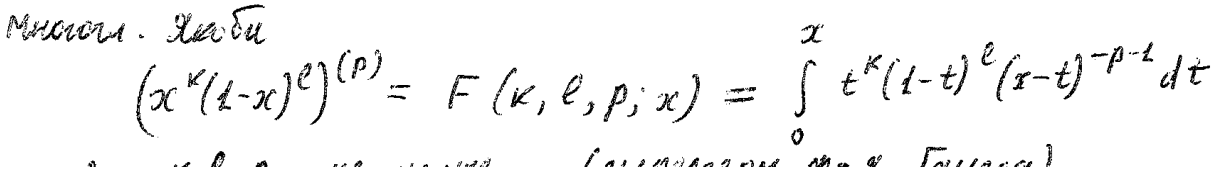


Diagram and typed description. of the linear representation associated with it.

Original source details: GELFAND-P001016

Jacobi polynomials

$$\left(x^k(1-x)^\ell\right)^{(p)} = F(k,\ell,p;x) = \int_0^x t^k(1-t)^\ell(x-t)^{-p-1}\,dt$$



Original source details: GELFAND-P001016

Uncertain reading. There is no prefactor between the equals sign and the integral.

All k, ℓ, p — are nonintegers (Gauss hypergeometric function)

Fractional derivative:

$$(x^n)^{(k)} = x^{n-k}\frac{n!}{(n-k)!} \qquad \text{or} \qquad \left(\frac{x^n}{n!}\right)^{(k)} = \frac{x^{n-k}}{(n-k)!}$$

and replace ! by Γ.

There is no satisfactory theory of hypergeometric functions of several variables (important for representations)

Atiyah: generalization of the statement that the points $\in$ a convex polygon, to the case where only the Cartan group acts.

Idea: recast all of representation theory in terms of the action of the Cartan group.

References: 1) Berezin, Gelfand 2) Atiyah 3) Guillemin, Sternberg 4) Khovanskii

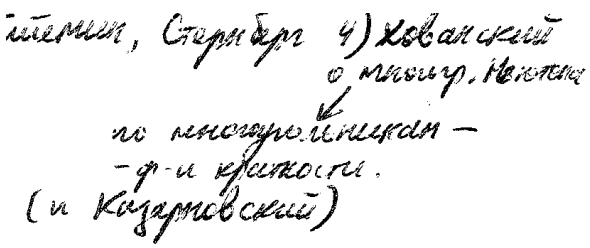


Diagram and typed description. on Newton polyhedra
↙
from polygons —
— multiplicity functions.
(and Kazarnovskii)

Original source details: GELFAND-P001016

GELFAND-P001017

[source number: 46]

29/III/82 Gelfand seminar.

Givental.

Thm. 1. M — a smooth manifold, N — a submanifold.

ω_0, ω_1 — two symplectic structures on M

$$\omega_0|_{TN} = \omega_1|_{TN}$$

Then $\forall x \in N$, $\exists$ a neighbourhood U and a diffeomorphism $g : U \to U$, such that

$$g|_N = \mathrm{id}_N, \qquad g^*\omega_1 = \omega_0.$$

— · —

A. Weinstein proved this if $\omega_0|_{TM|_N} = \omega_1|_{TM|_N}$. (he wrote $\omega_0|_N = \omega_1|_N$)

— · —

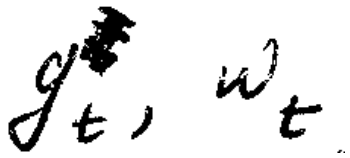
Moral: there is no extrinsic symplectic geometry for a submanifold.

— · —

There are also global formulations.

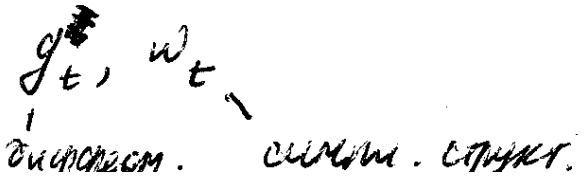
— · —

Idea of the proof: take a family g_t, ω_t



Original source details: GELFAND-P001017

Uncertain reading. Above g_t a mark is crossed out, possibly an asterisk.



Original source details: GELFAND-P001017

Diagram and typed description.

g_t — diffeomorphism

ω_t — symplectic structure

$$g_t^* \omega_t = \omega_0.$$

Differentiate and then use the existence and uniqueness theorem for differential equations. (reduces to a linear problem)

$N = \{*\} \Rightarrow$ a new proof of Darboux's theorem.

[Question: under what conditions is a 2-form obtained by restricting a symplectic form.

Uncertain reading. A short word is crossed out.



Original source details: GELFAND-P001017

necessary condition: $d\omega = 0$].

I. M. G. There is no need to add a closedness condition yet: a symplectic structure in $\forall$ tangent space.

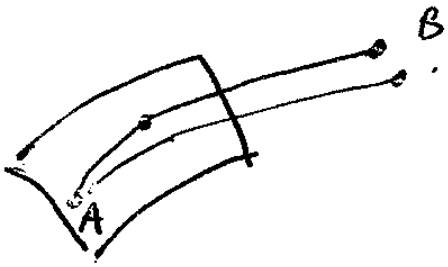
A symplectic form has no properties.

— · —

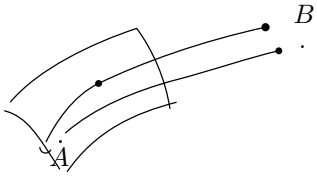
Symplectic structure in the space of polynomials.

($\Rightarrow$ Lagrangian manifolds with singularities)

Diagram and typed description.



Original source details: GELFAND-P001017



The small strokes at A leave the connection of the two curves unresolved.

A — a point on the surface.

B — an arbitrary point off the surface.

look for extremals (the geodesics break off)

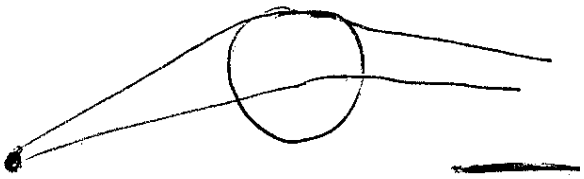
A. The space of lines in $\mathbb{R}^3$ has a natural symplectic structure!

B. Generalization: the space of extremals of any variational problem has a symplectic structure.

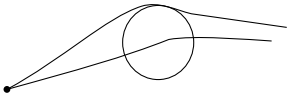
A. it is $T^*\mathbb{R}^2$ (Arnold's explanation)

Example:

Diagram and typed description.



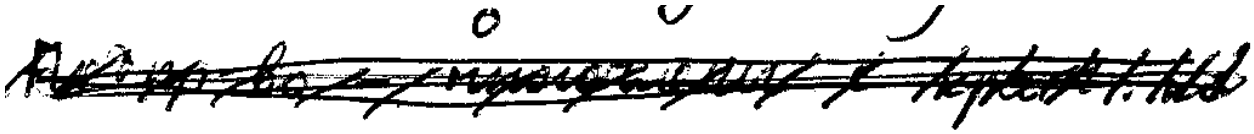
Original source details: GELFAND-P001017



a point and a sphere: a two-dimensional manifold of extremals arises.

— · —

Consider the space of binary forms $\{\sum_0^{2n-1} a_i x^i y^{2n-1-i}\}$ and on it the representation of the group $SL_2(\mathbb{R})$.



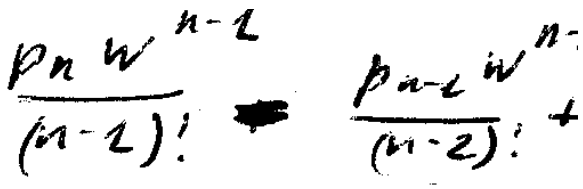
Original source details: GELFAND-P001017

Undeciphered passage. The sentence is crossed out. It may begin “This space” or “On the space”. Its full reading remains unresolved.

At $y = 1$ — simply a polynomial of degree $2n - 1$.

$w = \dfrac{x}{y}$ — an inhomogeneous coordinate. Write the polynomial in this variable using Darboux coordinates:

$$\frac{q_1 w^{2n-1}}{(2n-1)!} + \ldots + \frac{q_n w^n}{n!} + \frac{p_n w^{n-1}}{(n-1)!} \boxed{?} \frac{p_{n-1} w^{n-2}}{(n-2)!} + \ldots + (-1)^{n-1} \frac{p_1 w^0}{0!}$$



Uncertain reading. The sign before the term with p_{n-1} is overwritten. Two horizontal strokes and a lower vertical stroke are visible. The final sign remains unresolved.

Original source details: GELFAND-P001017

GELFAND-P001018

$$\omega = \sum_{i=1}^{n} dp_i \wedge dq_i$$

The action of $SL_2(\mathbb{R})$ preserves the form.

$w \mapsto w + t$ is a Hamiltonian flow; since it comes from $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$

Hamiltonian:

$$h = q_n^2/2 - q_{n-1}p_n - \ldots - q_1 p_2$$

q_1 — a first integral.

Consider the reduction $q_1 = 1$ and the space of characteristics on it.

(we also take $p_1 = 1$ — this is a convention)

Question of I. Goldsheid.

Consider a symplectic space with canonical basis $e_1, \ldots, e_n, f_1, \ldots, f_n$

Let L be a Lagrangian plane in it. Is there a (Lagrangian) plane L_1, spanned by vectors $\varphi_1, \ldots, \varphi_n$, where $\varphi_j = \alpha_j e_j + \beta_j f_j$, such that

$$\rho(L, L_1) \leq 1/\sqrt{2}$$

(it suffices to have $< 1 - \varepsilon$, where $\varepsilon > 0$ is independent of the dimension)

Here $\rho(L, L_1) = \|P_L - P_{L_1}\|$, where the (operator) norm is taken in the standard Euclidean metric in the basis $e_1, \dots, e_n, f_1, \dots, f_n$.

It is asserted that $\exists \varphi_j$ such that $\sin(\widehat{\varphi_j, L}) \le 1/\sqrt{2}$.

GELFAND-P001019

5/IV/82 Gelfand seminar.

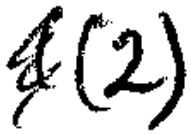
Wodzicki

M — a manifold (possibly with boundary ∂M)

D — a system of differential equations on M:

$$(1) \qquad D : C^\infty(M, E) \longrightarrow C^\infty(M, E)$$

elliptically determined (σ_D is invertible).



Uncertain reading. A mark before the equation number (2) is crossed out. Its reading remains unresolved.

Original source details: GELFAND-P001019

$$(2) \qquad \operatorname{index} D = \dim \ker D - \dim \operatorname{Coker} D = \int_M \alpha_0(x) \qquad \text{(Atiyah–Singer)}$$

[Below $\alpha_0(x)$: density.]

(2) — an example of a local formula.

The coefficient α_0 is expressed in terms of the symbol. Here there naturally appears a ζ-function

$$(3) \qquad \zeta_A(s) = \sum_{\lambda \in \operatorname{Spec} A} \lambda^{-s}$$

$$A = D^*D, \quad DD^*.$$

$$z_j = \frac{n-j}{m} \quad \text{— poles,} \qquad n = \dim M, \quad m = \text{order of } A.$$

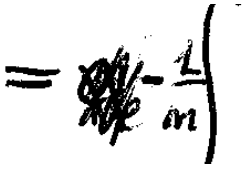
$$z_j = 0, -1, -2, \dots \quad \text{— there are no poles.}$$

$$\int_M \underbrace{\zeta_A(0; x)}_{\text{density}} = \zeta(0); \qquad \zeta_A(0; x) \text{ can be computed explicitly}$$

$$(4) \qquad \operatorname{index} D = \zeta_{1+D^*D}(0) - \zeta_{1+DD^*}(0)$$

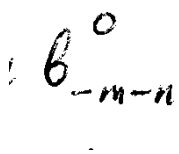
GELFAND-P001020

$$(5) \qquad \zeta_A(0; x) = -\frac{1}{m} \int_{|\xi|=1} d\xi' \int_0^\infty \operatorname{tr} b^0_{-m-n}(x, \xi, \lambda)\, d\lambda$$



Uncertain reading. A sign or factor is crossed out between the equals sign and $-1/m$.

Original source details: GELFAND-P001020



Uncertain reading. A small oval is written above b. It could be 0, o or $\circ$.

Original source details: GELFAND-P001020

[A line from b^0_{-m-n} to the explanation on the right:] component of the symbol of the parametrix for $(A-\lambda)$ in some chart.

$\zeta_A(0;x)\,dx$ — a density (!) — this is not obvious.

Suppose $\partial M \neq \varnothing$.

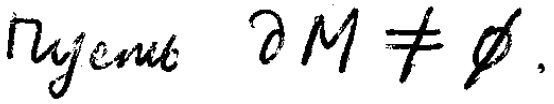


Diagram and typed description. The manuscript has $\partial M \neq \varnothing$.

Original source details: GELFAND-P001020

D — the signature operator (Dirac)

For this operator:

$$\text{(6)} \qquad \operatorname{index} D = \operatorname{sign} M$$

$$\text{(7)} \qquad \int_M L_k(p) = \operatorname{index} D = \int_M \alpha_0(x)$$

$$D : \Omega_+ \longrightarrow \Omega_-$$

$\Omega_\pm$ — forms even (odd) under the involution $\tau = i^{np+n+1}*$ (on p-forms).

$$\text{(8)} \qquad d + d^* = D|_{\Omega_+} : \Omega_+ \longrightarrow \Omega_-.$$

Example (I. M. G.)

$$\text{(9)} \qquad \begin{pmatrix} 0 & \operatorname{div} \\ \operatorname{grad} & \operatorname{rot} \end{pmatrix}^2 = \Delta_{\substack{I \\ \text{identity matrix}}} \qquad (\text{in } R^4)$$

$$\text{(10)} \qquad \text{Theorem.} \quad L_k(p) = \alpha_0(x) \qquad \text{(Atiyah–Bott–Patodi)}$$

$$n = 4 \quad \Rightarrow \quad \alpha_0(x) = \frac{1}{3} p_1$$

The case



Uncertain reading. Marks before D are crossed out.

Original source details: GELFAND-P001020

$D = \bar{\partial}$ and $\overset{\text{metric}}{M}$ is Kähler $\Rightarrow$ local agreement of $\alpha_0(\dots)$ and the component of the Todd genus $T(p)$ (in general this is false).



Original source details: GELFAND-P001020

Uncertain reading. The abbreviation before "Todd genus" most likely reads «соств.».

That is, there are two local formulas that do not coincide.

$\partial M \neq \varnothing \Rightarrow$ there is also a signature (degenerate form on middle cohomology, with zeros disregarded)

$$(11) \qquad \underset{\substack{\text{topological}\\ \text{invariant}}}{\operatorname{sign} M} = \underbrace{\int_M L_k(p)}_{\text{local formula}} - \underbrace{\eta(\partial M)}_{\substack{\text{depends}\\ \text{on the metric}\\ \text{at the boundary}}}$$

(the metric near the boundary — is a product).

I. M. G. Problem: is sign — the only additive invariant for manifolds



Original source details: GELFAND-P001020

Uncertain reading. A short word before "with" is crossed out.

with boundary?

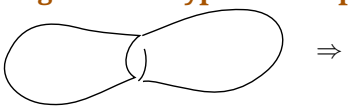
— · —

Pass to the double:



Original source details: GELFAND-P001020

Diagram and typed description.



There is a gap at the top of the right-hand stroke of the seam.

By Hirzebruch's theorem (equation (7)), we see that $\eta(\partial M)$ is independent of the filling.

(12) $\eta(\partial M)$ — a nonlocal invariant!!

Locality $\Rightarrow$ multiplicativity (with respect to [inserted above the line] passing to coverings)

There are examples (lens spaces) where this fails.

GELFAND-P001021

[source number: 58]

(13) Theorem. (Atiyah–Patodi–Singer)
$N = \partial M$, D — a first-order system

$$D = \frac{\partial}{\partial u} + A, \qquad A \text{ — an operator on } \partial M \tag{14}$$

(near the boundary)
Let A be normal with spectrum disjoint from a conical neighbourhood of $i\mathbb{R}$.

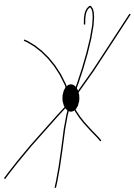


Uncertain reading. Between “not” and “intersecting” is a crossed-out abbreviation, possibly «сод.» or «соб.».

Original source details: GELFAND-P001021



Diagram and typed description.



Original source details: GELFAND-P001021

Then $\exists$ elliptic boundary conditions for D.
P_A — the projector onto the eigenvalues of A with Re $A \geq 0$.



Diagram and typed description. The condition reads Re $A \geq 0$.

Original source details: GELFAND-P001021

$$\left\{\begin{matrix} Df \\ P_A(f|_A) = 0 \end{matrix}\right\} \quad \text{— a nonlocal boundary-value problem} \tag{15}$$

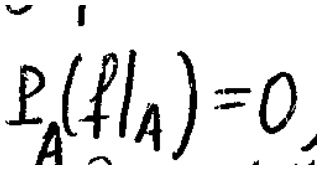


Diagram and typed description. The restriction subscript in (15) is written A.

Original source details: GELFAND-P001021

a local problem does not always exist.

But a nonlocal problem always exists!!



Original source details: GELFAND-P001021

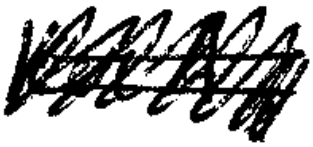
Uncertain reading. A mark after “nonlocal” is crossed out.

Then:

$$\operatorname{index} D = \int \alpha_0(x) - \frac{h_0 + \eta(A)}{2}, \tag{16}$$

where

$h_0 =$ dim (space of purely imaginary eigenvalues).



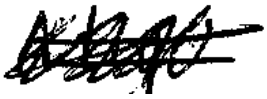
Original source details: GELFAND-P001021

Uncertain reading. The entry after dim is crossed out. It may read Ker A or Ker A_0.

$$\eta(A) \equiv \eta(0, A), \qquad \eta(s, A) = \sum_{\lambda \in \operatorname{Spec} A} \operatorname{sign} \operatorname{Re} \lambda \, |\operatorname{Re} \lambda|^{-s} \tag{17}$$

Point: generalization of the L-function.

In A.-P.-S. A — a self-adjoint operator

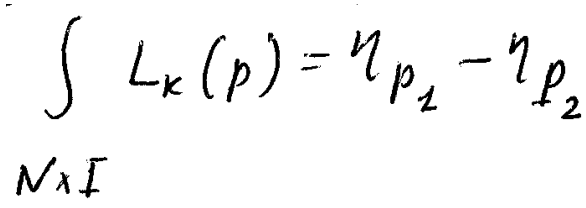


Original source details: GELFAND-P001021

Uncertain reading. The entry before A.-P.-S. is crossed out. It may read «Водз.» or «Водз».

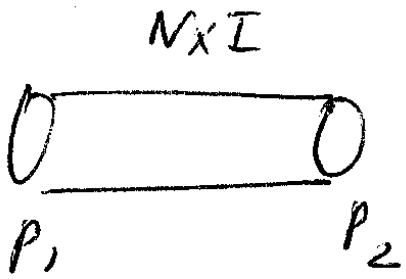
Theorem. A — an elliptic operator.

$$\int_{N \times I} L_k(p) = \eta_{p_1} - \eta_{p_2}$$



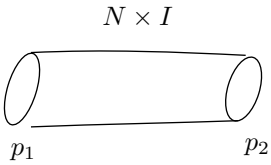
Original source details: GELFAND-P001021

Uncertain reading. The handwritten letters in $L_k(p)$ and the subscripts of η resemble p. The subscripts may be ρ.



Original source details: GELFAND-P001021

Diagram and typed description.



The end labels read p_1, p_2 or ρ_1, ρ_2.

GELFAND-P001022

12/IV/82 Gelfand seminar.

Wodzicki. (Mariusz)

X — a compact manifold with boundary.



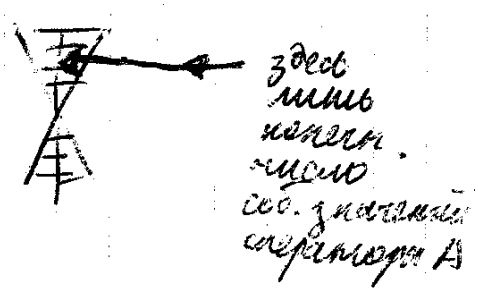
Original source details: GELFAND-P001022

Diagram and typed description. The manuscript abbreviates “manifold” as «мн-зие».

(1) $D\colon C^\infty(X,E) \to C^\infty(X,F)$ is a first-order elliptic system which (differential) near ∂X takes the form

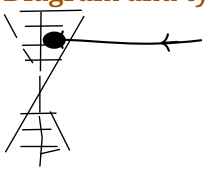
$$(2) \qquad D = \frac{\partial}{\partial u} + A, \qquad A \text{ — an operator on the boundary}$$

$$(3) \qquad A\colon C^\infty(Y, E|_Y) \longrightarrow C^\infty(Y, F|_Y)$$



Original source details: GELFAND-P001022

Diagram and typed description.



[Label with arrow:] only finitely many eigenvalues of the operator A.
Dense strokes at the tip obscure the exact joins.

Boundary conditions:

$$(4) \qquad P_+ f|_Y = 0,$$

P_+ — the projector onto the positive eigenfunctions of the operator A.

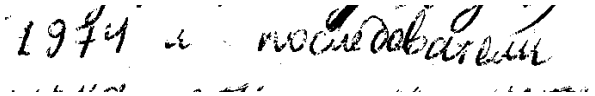


Original source details: GELFAND-P001022

Uncertain reading. A mark is crossed out between the words abbreviated as “eigenfunctions”.

(5) Theorem (Atiyah–Patody–Singer)

1974 and followers



Original source details: GELFAND-P001022

Uncertain reading. The small word after "1974" most likely reads "and".

suppose

1) A is normal with respect to the Riemannian metric on the boundary $\quad (A^*A = AA^*)$

$$(6) \qquad \operatorname{index} D = \int_X \alpha_0(x)\, dx - \frac{h(A) + \eta(A)}{2}$$

$$(7) \qquad \eta(A, s) = \sum \operatorname{sign} \operatorname{Re} \lambda \cdot |\operatorname{Re} \lambda|^{-s}, \qquad \eta(A) = \eta(A, 0)$$



Original source details: GELFAND-P001022

Uncertain reading. The number beside the formula for $\eta(A, s)$ is overwritten. The top stroke and the numbering sequence support (7).

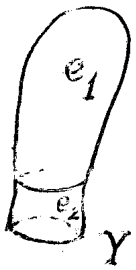
$$(8) \qquad h(A) = \dim \left\{ \begin{array}{c} \text{imaginary} \\ \text{eigenvalues of } A \end{array} \right\}$$

GELFAND-P001023

[source number: 64]

Proof. Use the formula:

$$(9) \qquad \operatorname{index} D = \underbrace{\operatorname{Tr} e^{-tD^*D}}_{\sum_{\lambda \in \operatorname{Spec} D^*D} e^{-t\lambda}} - \underbrace{\operatorname{Tr} e^{-tDD^*}}_{\sum_{\mu \in \operatorname{Spec} DD^*} e^{-t\mu}}$$



Original source details: GELFAND-P001023

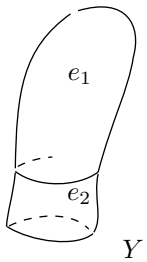


Diagram and typed description. The lower-right label reads Y. The rear arcs are broken.

— · —

A — a self-adjoint elliptic operator.



Uncertain reading. A word is crossed out before “elliptic”.

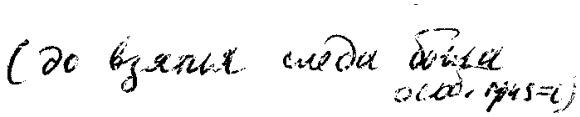
Original source details: GELFAND-P001023

$$(10) \qquad \sum_{\lambda>0} \lambda^{-s} = \zeta_+(s), \qquad \zeta_+(A) \text{ — the value at } s = 0.$$

(11) $\zeta_+(A) < \infty$, but locally there is a residue!! (nonlocal functional)

$$(12) \qquad \zeta_+(s) = \mathrm{Tr}(P_+ A^{-s}) = \int_X \mathrm{Tr}(P_+ A^{-s})(x,x)$$

[At right: (there were singularities before taking the trace at $s = i$).]



Uncertain reading. The sign after $s =$ resembles i, with a faint separate stroke above it. The reading remains uncertain.

Original source details: GELFAND-P001023

The local term has a residue, and $\int_X$ of this residue $= 0$.



Uncertain reading. The abbreviation after “local” most likely means “term”. The last strokes are faint.

Original source details: GELFAND-P001023

$$(13) \qquad \begin{cases} \zeta(A) = \zeta_+(A) + \zeta_-(A) \\ \eta(A) = \zeta_+(A) - \zeta_-(A) \end{cases}$$

I. M. G.: in the work of Gabrielov, Gelfand and Fuks there are algebraic reasons for nonlocality.

The most remarkable problem is unsolved: this is done to compute the values at the boundary of the forms of Pontryagin, but for general classes there is nothing.

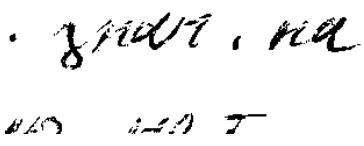


Diagram and typed description. The manuscript abbreviates “values” as «знач.» before “at the boundary”.

Original source details: GELFAND-P001023

Stredder Epstein (2 papers)	J. Diff. Geometry 10 (1975), 645	on natural differential operators.

— · —

Question of I. M. G.: why require self-adjointness of A?
Let a domain and an operator on it be given, the operator A.



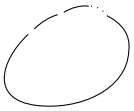


Diagram and typed description.

Original source details: GELFAND-P001023

One constructs a theory of cobordisms.
The invariant [below it: $\eta(A)$] A is constructed with a membrane stretched over the manifold.

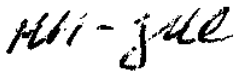


Diagram and typed description. The manuscript abbreviates “manifold” as «мн-зие».

Original source details: GELFAND-P001023

— · —

But in the absence of a filling this has yet to be proved!!

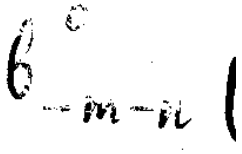
— · —

$$\alpha_0(x) = \zeta(D^*D + \varepsilon, x) - \zeta(DD^* + \varepsilon, x) \tag{14}$$

GELFAND-P001024

Formula for $\zeta(A, x)$

$$\frac{1}{m} \int_{|\xi|=1} d\xi' \int_0^\infty b^\circ_{-m-n}(x, \xi, \lambda)\, d\lambda \tag{15}$$



Uncertain reading. In (15), a small broken oval above b is rendered as a circle.

Original source details: GELFAND-P001024

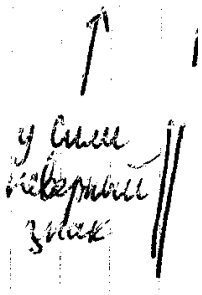


Diagram and typed description.
↑
Seeley has the wrong sign‖

Original source details: GELFAND-P001024

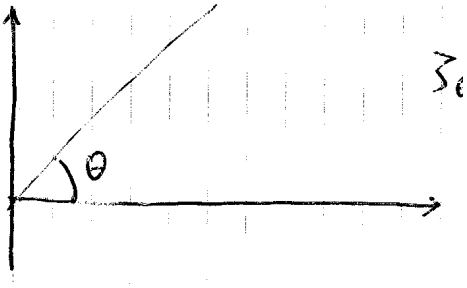
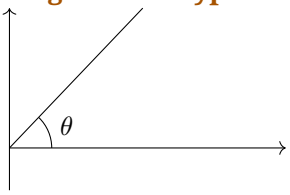


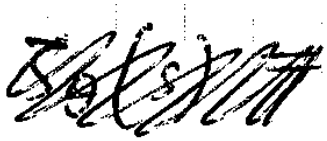
Diagram and typed description.



Original source details: GELFAND-P001024

$$\zeta_\theta(s) = \sum \lambda^{-s}$$

(with a cut along the ray at angle θ)

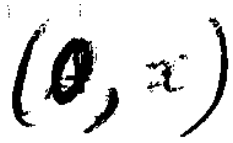


Uncertain reading. The crossed-out entry before (16) appears to read $\zeta_\theta(s) =$.

Original source details: GELFAND-P001024

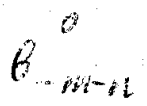
$$\zeta_\theta^A(0,x) = -\frac{e^{i\theta}}{m}\int_{|\xi|=1} d\xi' \int_0^\infty b^\circ_{-m-n}(x,\xi,e^{i\theta}\lambda)\,d\lambda \tag{16}$$

(this is the correct formula)



Uncertain reading. The first argument in (16) is overwritten. Both 0 and θ are possible.

Original source details: GELFAND-P001024



Uncertain reading. In (16), the superscript oval on b is rendered as a circle.

Original source details: GELFAND-P001024

I. M. G. A remarkable operator in $\mathbb{R}^3$ (elliptic):

$$\mathcal{A}\varphi = \begin{pmatrix} 0 & \operatorname{div} \\ \operatorname{grad} & \operatorname{rot} \end{pmatrix} \begin{pmatrix} \varphi_0 \\ \vec{\varphi} \end{pmatrix} = (\operatorname{div}\vec{\varphi}, \operatorname{grad}\varphi_0 + \operatorname{rot}\vec{\varphi}) \tag{17}$$

$$\mathcal{A}^2 = \begin{pmatrix} \Delta & 0 \\ 0 & \Delta \end{pmatrix}.$$

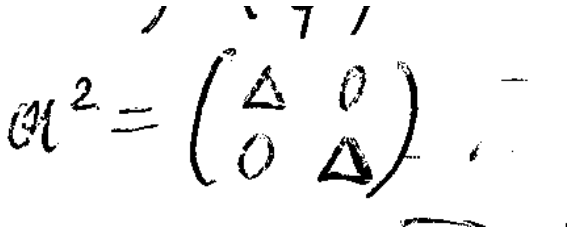


Diagram and typed description. Both diagonal entries read Δ.

Original source details: GELFAND-P001024

— · —

A — an elliptic operator on a closed manifold, the spectrum of A does not fill a sector around the ray (ray of min growth)



Uncertain reading. The word most likely means “fill”. “Meet” is also possible.

Original source details: GELFAND-P001024

$\underbrace{\zeta_{\theta,A}(s) = \sum \lambda_\theta^{-s}}$ define using the ray.

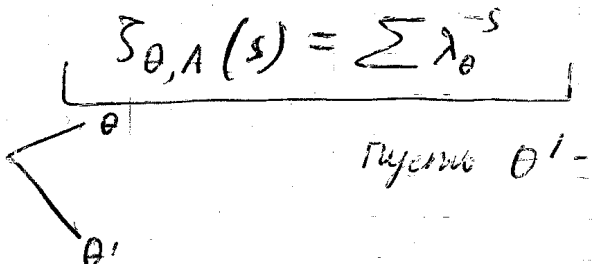
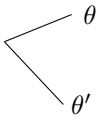


Diagram and typed description.

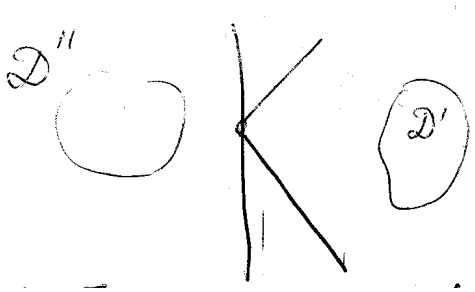


Original source details: GELFAND-P001024

Let θ' — be another ray of min growth.

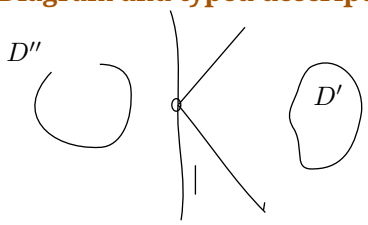
$$\zeta_{\theta',A}(s) = \sum \lambda_{\theta'}^{-s}$$

$\Rightarrow$ the spectrum lies in two sectors



Original source details: GELFAND-P001024

Diagram and typed description.



The left contour has a gap. The tick at the bottom of the slanted stroke may be part of an arrow.

(18) Theorem.

$$\rho(s) \stackrel{\text{def}}{=} \zeta_\theta(A,s) - \zeta_{\theta'}(A,s)$$

Then

a) $\rho(s)$ is meromorphic with residues at $z_j = \dfrac{-j+n}{m}$, $j = 0,1,2,\ldots$

b) $z_j \in \mathbb{Z} \Rightarrow$ the residue vanishes microlocally!!

(elementary!)

GELFAND-P001025

[source number: 65]

c)

$$\begin{array}{ccc} z_j & \in \mathbb{Z} \Rightarrow & \rho(z_j) \\ \| & & \| \\ \ell & & \rho(\ell) \end{array} \quad \text{is given by local formulas}$$

(a substantive assertion, — also valid for pseudodifferential operators!)

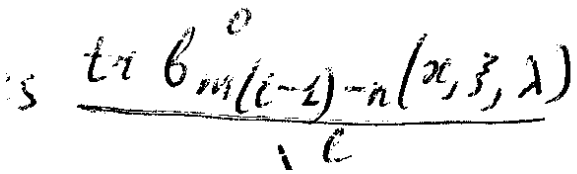
$$(19) \qquad \rho(i) \stackrel{\text{def}}{=} \int_X \rho_\ell(x)\,dx$$

$$= -\frac{2\pi i}{m} \int_X dx \int_{|\xi|=1} d\xi' \sum_{\lambda_\nu \in D'} \operatorname{Res} \frac{\operatorname{tr} b^\circ_{m(i-1)-n}(x,\xi,\lambda)}{\lambda^\ell}$$



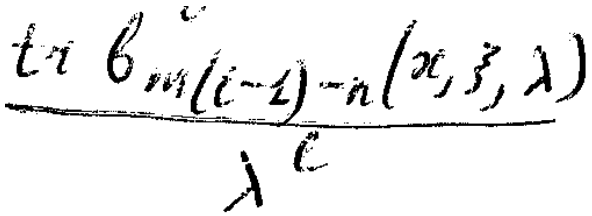
Original source details: GELFAND-P001025

Uncertain reading. The argument of $\rho(i)$ resembles i, and the subscript of $\rho_\ell(x)$ resembles ℓ. Part (c) uses ℓ.



Original source details: GELFAND-P001025

Uncertain reading. A broken oval above b is rendered as a circle.



Original source details: GELFAND-P001025

Uncertain reading. The index of b appears to contain i, and the exponent of λ appears to be ℓ.

d) $\ell \in \mathbb{Z}_- = \{0, -1, -2, -3, -4, \ldots\} \Rightarrow \rho(\ell) = 0$

(a stronger version of A.-P.-S.).

Hints (for (d)):

The proof should cover more than just the residues at 0, and should cover all integer points > 0



Original source details: GELFAND-P001025

Uncertain reading. The word after "about" most likely reads "residues".

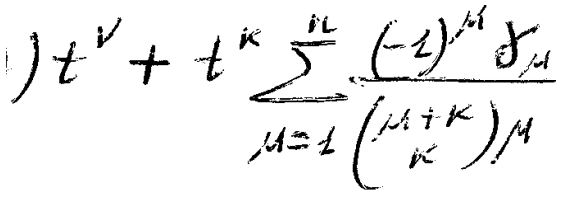


Original source details: GELFAND-P001025

Diagram and typed description. The inequality in the hint reads > 0.

At the point 0 it works out: use the Mahler transform (from the p-adic numbers):

$$\zeta_\theta(-k, A+t) = \underbrace{\sum_{\nu=0}^{k} \binom{k}{\nu} \zeta(\nu - k, A) t^\nu}_{\text{polynomial:}} + t^k \sum_{\mu=1}^{n} \frac{(-1)^\mu \delta_\mu}{\binom{\mu+k}{k} \mu} \tag{20}$$



Original source details: GELFAND-P001025

Uncertain reading. The second sum is preceded by t^k. No further power of t is visible in the summand.

δ_μ — the residue at the point μ for the ζ-function of the operator A
This is for first-order operators.

$$\text{Ker } A \neq \{0\} \Rightarrow \text{instead of } \zeta(0) \text{ write } \zeta(0) + h_0(A)$$

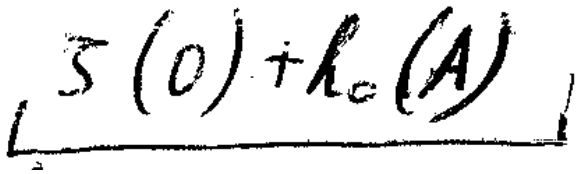
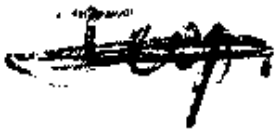


Diagram and typed description. $\zeta(0) + h_0(A)$

Original source details: GELFAND-P001025

($h_0(A) = \dim \text{Ker } A$).



Uncertain reading. The entry before “first sum” is crossed out and may read “Thm.”

Original source details: GELFAND-P001025

The first sum in (20) — the Mahler transform.
An automorphic function with respect to the Mahler transform extends to the p-adic domain.